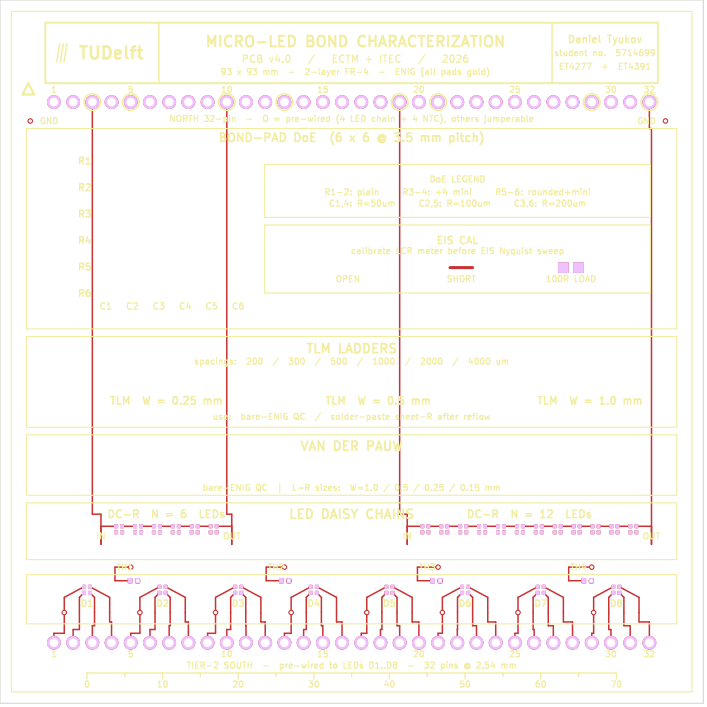
TUDelft
MICRO-LED BOND CHARACTERIZATION
PCB v4.0 / ECTM + ITEC / 2026
93 x 93 mm – 2-layer FR-4 – ENIG (all pads gold)
Daniel Tyukov
student no. 5714699
ET4277 + ET4391
NORTH 32-pin – 0 = pre-wired (4 LED chain + 4 NTC), others jumperable
GND
BOND-PAD DoE (6 x 6 @ 3.5 mm pitch)
R1
R2
R3
R4
R5
R6
C1 C2 C3 C4 C5 C6
DoE LEGEND
R1–2: plain R3–4: +4 mini R5–6: rounded+mini
C1,4: R=50um C2,5: R=100um C3,6: R=200um
EIS CAL
calibrate LCR meter before EIS Nyquist sweep
OPEN
SHORT
100R LOAD
TLM LADDERS
spacings: 200 / 300 / 500 / 1000 / 2000 / 4000 um
TLM W = 0.25 mm
TLM W = 0.5 mm
TLM W = 1.0 mm
use: bare-ENIG QC / solder-paste sheet-R after reflow
VAN DER PAUW
bare-ENIG QC | L=R sizes: W=1.0 / 0.5 / 0.25 / 0.15 mm
LED DAISY CHAINS
DC-R N = 6 LEDs
DC-R N = 12 LEDs
IN
OUT
TIER-2 SOUTH – pre-wired to LEDs D1..D8 – 32 pins @ 2.54 mm